Multi-Peak Ground-State Solutions for the Nonlinear Schrödinger Equation with External Potential

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Abstract

We classify multi-peak ground-state configurations of the stationary nonlinear Schrödinger equation $(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0$ in the high-energy regime $E \to \infty$. Using the rescaling and concentration compactness framework of Kirr and Natarajan [10], ground states decompose into well-separated profiles whose positions are governed by a finite-dimensional algebraic system balancing the Hessian $H_V(x_0)$ of the potential against nonlinear inter-peak interactions.

We prove that near a non-degenerate local maximum x_0 of V, the admissible multi-peak configurations are: collinear arrangements along a single eigenvector of H_V; isosceles triangles requiring the eigenvalue constraint $\lambda_2 = 3\lambda_1$; equilateral triangles requiring $|\lambda_2|/|\lambda_1| \geq 3$; and equilateral triangles with a continuous rotational degree of freedom θ, admissible for all orientations when $\frac{1}{3} \leq \mu \leq 3$ (where $\mu = |\lambda_2|/|\lambda_1|$) and for restricted angular windows otherwise. To extend these to finite energy, we construct explicit solutions of the perturbation system $F_i(\delta y, R) = 0$ at $R = 0$, identify the four-dimensional kernel of the linearization, and show that gauge-fixing constraints reduce the problem to a system amenable to the implicit function theorem.

Keywords: nonlinear Schrödinger equation, multi-peak solutions, concentration compactness, Lyapunov–Schmidt reduction, ground states, optical fibers

GRANT PLACEHOLDER

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1 Introduction

The propagation of intense light through optical fibers is governed by the interplay between linear dispersion and the nonlinear response of the dielectric medium. In silica, the centrosymmetric lattice eliminates second-order nonlinearity, so the leading nonlinear effect is the Kerr term $\chi^{(3)}$. Starting from Maxwell's equations in a source-free, non-magnetic dielectric, the electric field can be written as a slowly varying envelope modulating a carrier:

$$\mathbf{E}(\mathbf{r},t) = \tfrac{1}{2}\hat{\mathbf{x}}\big[F(x,y)\,A(z,t)\,e^{i(\beta_0 z-\omega_0 t)} + \text{c.c.}\big].$$

Here $F(x,y)$ is the transverse modal shape set by the fiber's refractive-index profile, $A(z,t)$ is the envelope, and β_0 is the propagation constant at the carrier frequency ω_0. Applying the slowly varying envelope approximation and expanding the propagation constant to second order in frequency yields the nonlinear Schrödinger equation (NLS):

$$i\partial_z A - \tfrac{\beta_2}{2}\partial_T^2 A + \gamma|A|^2 A = 0,$$

where β_2 is the group-velocity-dispersion parameter and $\gamma = n_2\omega_0/(cA_{\text{eff}})$ encodes the Kerr nonlinearity. The existence of optical solitons in this setting was predicted by Hasegawa and Tappert [5, 6] for both anomalous and normal dispersion regimes.

In real fibers the refractive index is not spatially uniform. Fiber Bragg gratings impose periodic modulations, graded-index fibers create a parabolic transverse profile (an effective harmonic-oscillator potential), and photonic-crystal fibers achieve confinement via a micro-structured air-hole lattice. In every case, the spatial variation of the index enters the governing equations as an external potential $V(\mathbf{x})$. The transverse modal equation

$$\nabla_\perp^2 F + \Big[\big(\tfrac{n\omega}{c}\big)^2(\mathbf{x}_\perp) - \beta^2\Big]F = 0$$

is mathematically identical to the stationary Schrödinger equation $-\Delta\psi + V(x)\psi = E\psi$. Guided modes of the fiber correspond to bound states of this effective potential, and the fundamental mode is its ground state. This equivalence motivates the present work: engineering $V(\mathbf{x})$ via refractive-index design directly translates into questions about the existence, uniqueness, and stability of ground-state solutions of the NLS with an external potential.

Consequently, we study the stationary NLS

$$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0, \tag{1.1}$$

with $V : \mathbb{R}^n \to \mathbb{R}$, $\psi : \mathbb{R}^n \to \mathbb{C}$, $\sigma > 0$, and $E \geq 0$. The parameter E represents the energy (or, in the optical setting, the propagation constant). For each admissible E we seek ground-state solutions ψ_E and analyze how their spatial structure is governed by the geometry of V.

Since (1.1) is posed on the unbounded domain $\mathbb{R}^n$, compactness is lost and standard variational methods fail. The concentration compactness principle introduced by Lions [13] classifies minimizing sequences into compactness, vanishing, or dichotomy. Using the rescaling of Kirr and Natarajan [10], one obtains a bounded family in H^1. Concentration compactness then yields a multi-peak decomposition: u_E is approximated by a finite sum of well-separated profiles, each solving the translation-invariant NLS. The rescaled peak positions converge to the critical points of V as $E \to \infty$.

This multi-peak decomposition reduces the infinite-dimensional problem to a finite-dimensional system governing the peak locations. A Lyapunov–Schmidt reduction first solves for the orthogonal remainder and then projects onto the span of the translational modes, yielding an algebraic system that balances the restoring force from $H_V(x_0)$ against the attractive interaction between neighboring peaks. The overall analytical pipeline, building on [10] and extended in the present work, is illustrated in Figure 1.1.

Our contributions are threefold. First, we systematically classify all admissible multi-peak configurations near a non-degenerate local maximum of V (Theorem 2). Second, we formulate and solve the perturbation system at $R = 0$ (Proposition 3), identify the kernel obstruction (Proposition 4), and propose a gauge-fixing strategy for its resolution (Section 5). Third, we provide explicit admissibility regions in the (μ, θ) parameter space for equilateral triangle configurations with rotational freedom.

The multi-peak construction builds on a long line of work on semiclassical and spike solutions of NLS. Floer and Weinstein [2] first proved the existence of standing waves concentrated near nondegenerate critical points of V for bounded potentials. The existence of ground states for nonlinear scalar field equations

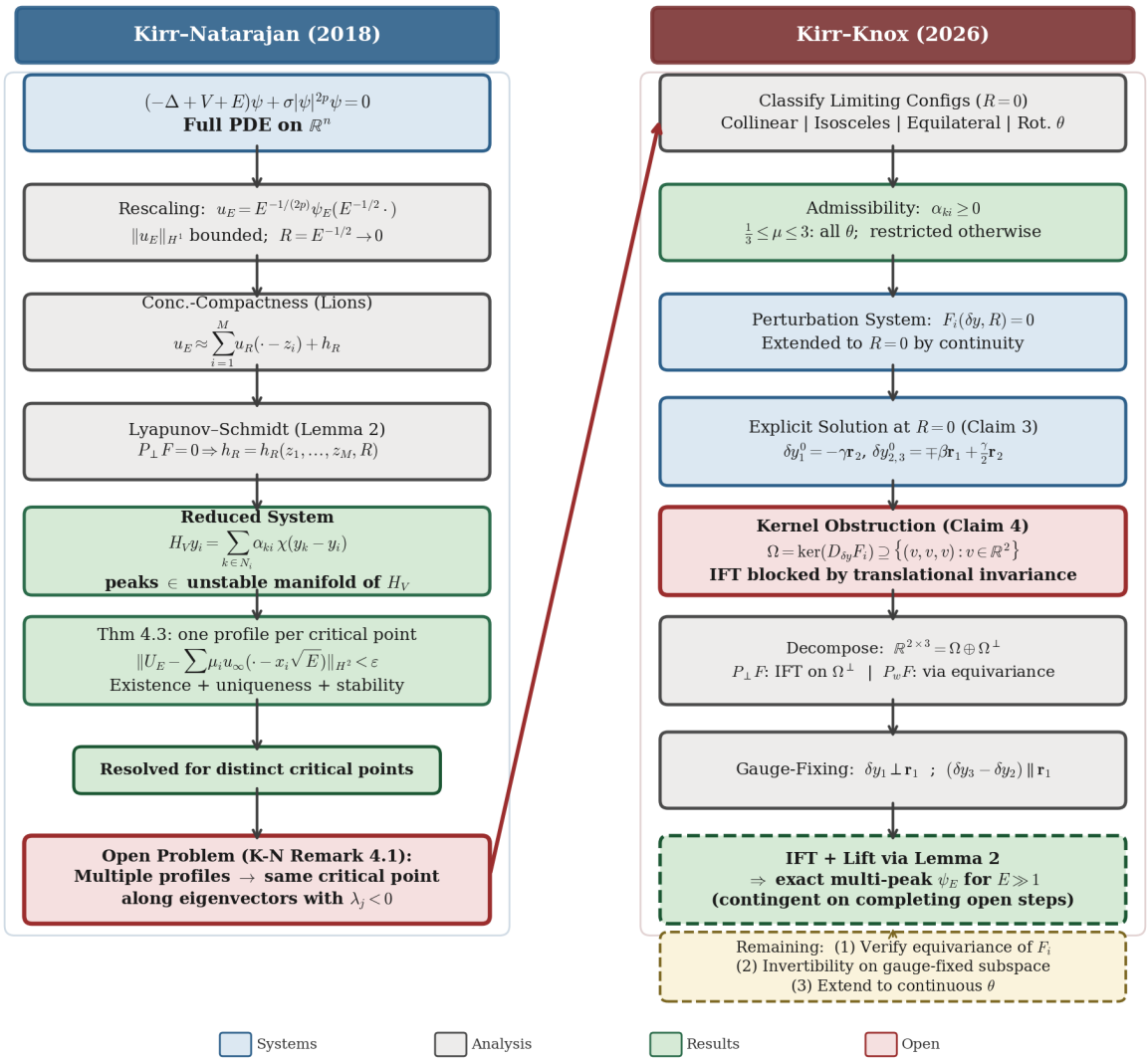

Figure 1.1: Complete analytical pipeline from the full PDE to the existence of multi-peak ground states. The top portion (marked "Kirr-Natarajan, 2018") covers the rescaling, concentration-compactness decomposition, and Lyapunov-Schmidt reduction established in [10]. The lower portion (marked "This work") covers the classification of admissible configurations, the perturbation system $F_i(\delta y, R) = 0$, the kernel obstruction and its resolution via gauge-fixing, and the application of the implicit function theorem on the constrained subspace. The open step (verifying rotational equivariance and invertibility on the complement) is indicated at left.

was established by Berestycki and Lions [1], and the concentration compactness framework for multi-peak decomposition was developed in [13, 14], and the symmetry-breaking bifurcation analysis for NLS with symmetric potentials was carried out in [8]. Stability of solitary waves under symmetry was established by Grillakis, Shatah, and Strauss [3, 4], and asymptotic stability of ground states was proved in [11, 9]. The present work extends the global bifurcation analysis of [10] by providing the first complete geometric classification of multi-peak configurations and a systematic resolution strategy for the kernel obstruction in the reduced system.

2 Setup and Preliminaries

2.1 Hypotheses on the potential

We impose the following conditions on V, following [8, 2, 10]:

Hypothesis V1. $V \in L^\infty(\mathbb{R}^n)$, i.e., V is bounded and measurable.

Hypothesis V2. $\lim_{|x|\to\infty} V(x) = 0$.

Hypothesis V3. V is C^2 in a neighborhood of x_0, where x_0 is a non-degenerate critical point of V: $\nabla V(x_0) = 0$ and $H_V(x_0)$ is invertible.

Hypothesis V4. All eigenvalues of $H_V(x_0)$ are strictly negative, i.e., x_0 is a strict local maximum of V. We write $\lambda_1, \dots, \lambda_n < 0$ for the eigenvalues with corresponding orthonormal eigenvectors $\boldsymbol{r}_1, \dots, \boldsymbol{r}_n$.

Hypothesis V1 ensures that $L_0 = -\Delta + V$ is self-adjoint on $L^2(\mathbb{R}^n)$ with domain $H^2(\mathbb{R}^n)$. Together with V2, it guarantees that the essential spectrum of L_0 is $[0, \infty)$ and that any negative eigenvalues are discrete. Hypothesis V3 is the minimal regularity needed for the Taylor expansion underlying the Lyapunov–Schmidt reduction. Hypothesis V4 is the structural condition that forces multi-peak configurations to concentrate at x_0; peaks at a local minimum would correspond to $\lambda_i > 0$, for which the algebraic system has no admissible solutions with $\alpha_{ki} \geq 0$.

Remark 1. Floer and Weinstein [2] work with V bounded and prove concentration near any nondegenerate critical point (maximum or minimum). For the multi-peak case, the requirement $\lambda_i < 0$ arises from the interaction analysis: the peaks must lie in the unstable subspace of $H_V(x_0)$ (Corollary 1).

2.2 Rescaling and concentration compactness

Following [10], define

$$u_E(x) = E^{-\frac{1}{2p}}\, \psi_E(E^{-\frac{1}{2}} x), \tag{2.1}$$

so that $\|u_E\|_{H^1}^2 = \|\nabla \psi_E\|_{L^2}^2 + \|\psi_E\|_{L^2}^2$ remains bounded. By the concentration compactness principle [13], there exists $M \in \mathbb{N}$ such that

$$\left\| u_E(x) - \sum_{i=1}^{M} u_i(x - z_i(E)) \right\|_{H^1} \xrightarrow{E\to\infty} 0, \tag{2.2}$$

where each profile $u_i : \mathbb{R}^n \to \mathbb{C}$ solves the translation-invariant equation

$$(-\Delta + 1)u + \sigma |u|^{2p} u = 0. \tag{2.3}$$

Setting $R = E^{-1/2}$, the rescaled NLS becomes

$$(-\Delta + R^2 V(Rx) + 1)\, u_E + \sigma |u_E|^{2p} u_E = 0. \tag{2.4}$$

Under Hypotheses V1–V2, the rescaled peak positions $Rz_i(R)$ remain bounded and converge to the critical points of V as $R \to 0$.

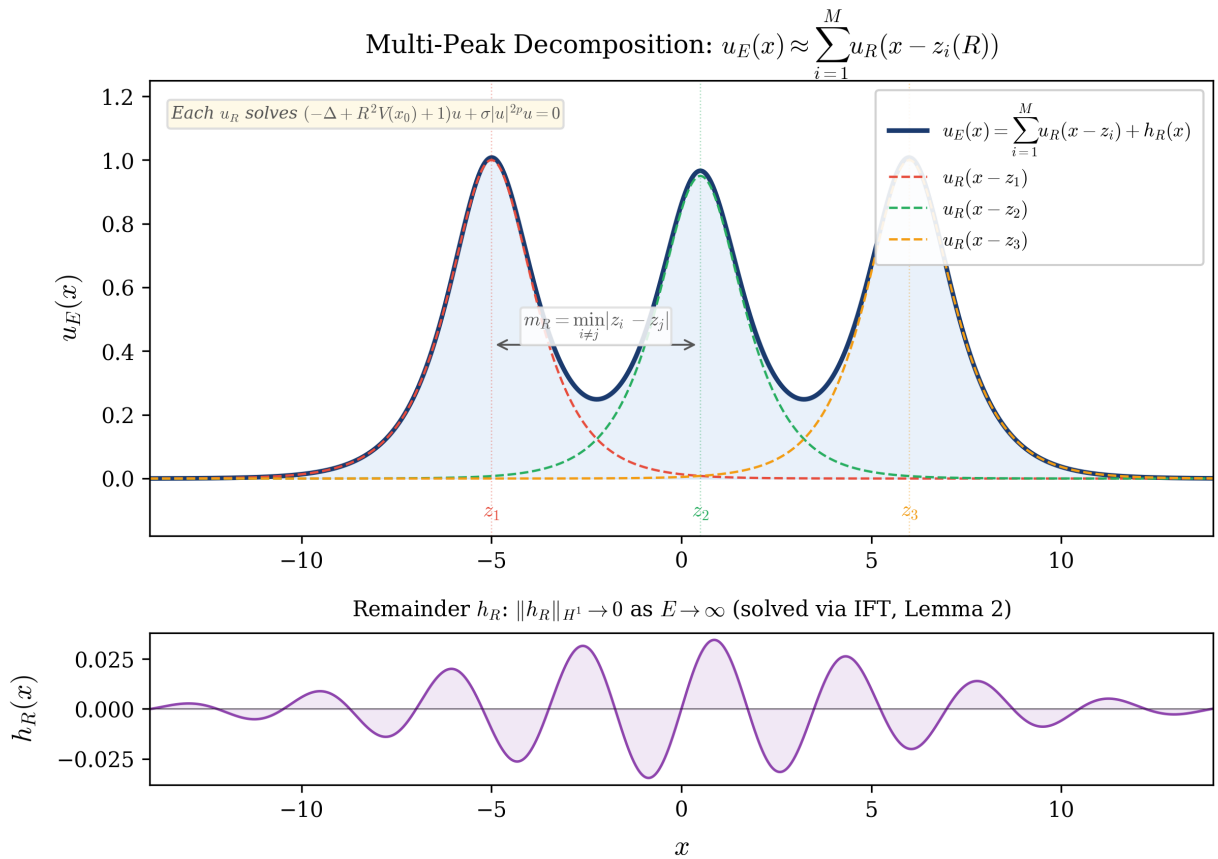

Figure 2.1: Illustration of the concentration-compactness decomposition (2.3) for $M = 3$ peaks.
Top: The rescaled solution $u_E(x)$ (solid blue) is approximated by a sum of well-separated profiles $u_R(x - z_i)$ (dashed), each solving $\left(-\Delta + R^2 V(x_0) + 1\right) u + \sigma |u|^{2p} u = 0$. The minimum inter-peak distance $m_R = \min_{i \neq j} |z_i - z_j|$ is indicated.
Bottom: The remainder $h_R(x)$, which satisfies $\|h_R\|_{H^1} \to 0$ as $E \to \infty$ and is recovered uniquely from the peak positions via the implicit function theorem (2).

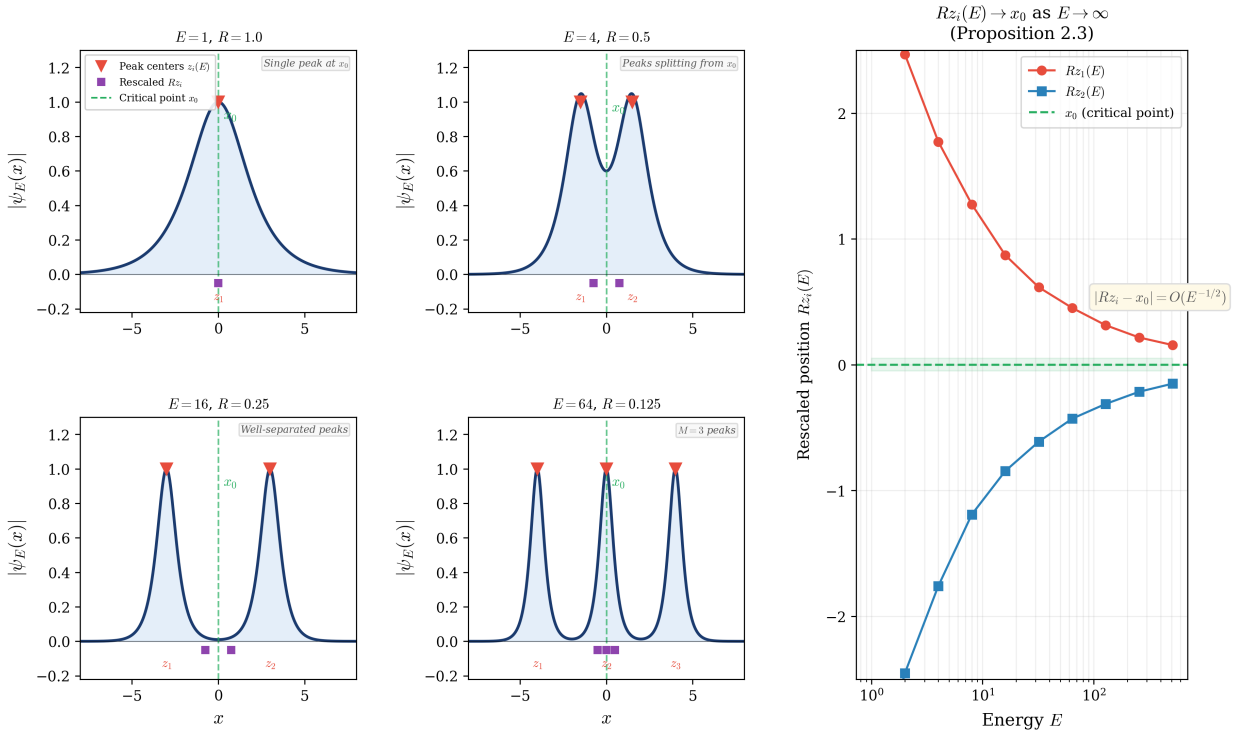

Figure 2.2: Evolution of peak structure with increasing energy E.
Left panels: Profiles $|\psi_E(x)|$ at $E = 1, 4, 16, 64$, showing the transition from a single peak to well-separated multi-peak structure. Red triangles mark the peak centers $z_i(E)$; purple squares mark the rescaled positions Rz_i converging toward the critical point x_0 (green dashed line).
Right: Convergence of $Rz_i(E) = E^{-\frac{1}{2}} z_i(E)$ to x_0 as $E \to \infty$, illustrating Claim 2. The convergence rate is $|Rz_i - x_0| = O\left(E^{-\frac{1}{2}}\right)$.

Proposition 1 (Kirr–Natarajan [10]). *Under Hypotheses V1–V4, $Rz_i(R) \to x_0$ for each $i = 1, \dots, M$ as $R \to 0$.*

We rewrite (2.2) as

$$u_E(x) = \sum_{i=1}^{M} u_R(x - z_i(R)) + h_R(x),$$

where u_R is the unique (up to phase) positive, spherically symmetric solution of $(-\Delta + R^2 V(x_0) + 1)u + \sigma|u|^{2p}u = 0$.

2.3 Lyapunov–Schmidt reduction

The derivatives $\partial u_R / \partial x_j$ satisfy the linearized equation and generate the translational kernel. The Lyapunov-Schmidt reduction proceeds in two steps:

Lemma 1 (Lyapunov–Schmidt, Step 1 [10]). *There exists $\varepsilon > 0$ such that for $0 < R < \varepsilon$, there exist adjusted centers $\tilde{z}_1, \dots, \tilde{z}_M \in \mathbb{R}^n$ with $u_E(x) = \sum_{i=1}^{M} u_R(x - z_i(R)) + \tilde{h}_R(x)$ and $\tilde{h}_R \perp \partial_{x_j} u_R(\cdot - z_i(R))$ for all $i = 1, \dots, M$ and $j = 1, \dots, n$.*

Lemma 2 (Lyapunov–Schmidt, Step 2 [10]). *There exist $\tilde{\varepsilon}, \delta, d > 0$ such that if $R < \tilde{\varepsilon}$, $\|h\|_{H^1} < \delta$, and $|z_i - z_j| > d$ for $i \neq j$, then $P_\perp F(h, z_1, \dots, z_M, R) = 0$ has a unique solution $h = h(z_1, \dots, z_M, R)$.*

2.4 The reduced algebraic system

Projecting onto the translational modes yields the reduced system. We introduce the following quantities:

Definition 1 (Interaction functional and coefficients). The *interaction functional* is

$$Q_R(z) = -\sigma \left\langle \partial_{z/|z|} u_R(\cdot - z),\, u_R^{2p+1} \right\rangle_{L^2}.$$

The *minimum inter-peak distance* is $m_R = \min\{|z_j(R) - z_i(R)| : i \neq j\}$. The *interaction coefficients* are

$$\alpha_{ki} = \frac{Q_R(|z_k - z_i|)}{Q_R(m_R)}, \quad \text{satisfying } 0 \le \alpha_{ki} \le 1, \tag{2.5}$$

with $\alpha_{ki} = 1$ precisely when $|z_k - z_i| = m_R$ (nearest neighbors).

Definition 2 (Overall interaction strength). The *overall interaction strength* is

$$\chi = \frac{2\, Q_R(m_R)}{\|u_R\|_{L^2}^2\; R^4\, m_R} > 0. \tag{2.6}$$

This is a single positive scalar that captures the magnitude of the inter-peak interaction relative to the Hessian restoring force. It is *not* equal to any individual α_{ki}; rather, the force on peak i from peak k has magnitude $\alpha_{ki}\chi$.

The orthogonality condition from the Lyapunov–Schmidt projection yields, after the rescaling $s_i = Rz_i - x_0$ and passage to the limit $R \to 0$ with $y_i = \lim_{R\to 0} s_i/(Rm_R)$:

$$H_V(x_0)\, y_i = \chi \sum_{k \in N_i} \alpha_{ki}(y_k - y_i), \tag{2.7}$$

where $N_i = \{k : |y_k - y_i| = 1\}$ is the set of nearest neighbors and we have used $|y_k - y_i| = 1$ for nearest neighbors.

Corollary 1. *The peak positions satisfy* $\operatorname{span}\{y_1, \dots, y_M\} \subseteq U$, *where*

$$U = \operatorname{span}\big\{\boldsymbol{r} \in \mathbb{R}^n : H_V(x_0)\boldsymbol{r} = \lambda \boldsymbol{r},\ \lambda < 0\big\}$$

is the unstable subspace of $H_V(x_0)$.

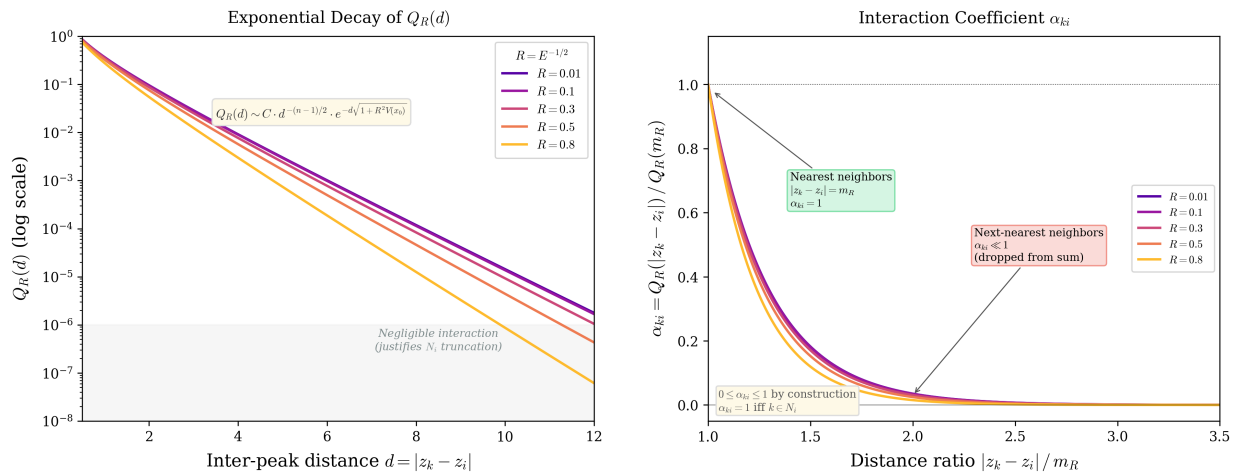

Figure 2.3: **(a)** Semilog plot of the interaction functional $Q_R(d) = -\sigma \left\langle \partial_{\hat{d}} u_R(\cdot - d), u_R^{2p+1} \right\rangle$ as a function of inter-peak distance d for several values of $R = E^{-\frac{1}{2}}$. The exponential decay $Q_R(d) \sim C \cdot d^{-\frac{n-1}{2}} \cdot e^{-d\sqrt{1+R^2V(x_0)}}$ ensures that only nearest-neighbor interactions contribute at leading order, justifying the truncation of the sum to $k \in N_i$.
(b) The interaction coefficient $\alpha_{ki} = \frac{Q_R(|z_k - z_i|)}{Q_R(m_R)}$ as a function of the distance ratio $\frac{|z_k - z_i|}{m_R}$. By construction $0 \le \alpha_{ki} \le 1$, with $\alpha_{ki} = 1$ for nearest neighbors ($|z_k - z_i| = m_R$) and $\alpha_{ki} \ll 1$ for next-nearest neighbors, which are dropped from the reduced system.

Proof. Write $y_i = y_i^U + y_i^S$ with $y_i^U \in U$ and $y_i^S \in U^\perp$. Applying the orthogonal projection P_S onto $U^\perp$ to the algebraic system (2.4) gives

$$H_V\, y_i^S = \chi \sum_{k \in N_i} \alpha_{ki} (y_k^S - y_i^S), \tag{2.8}$$

since H_V preserves both U and $U^\perp$ (they are eigenspaces) and $P_S(y_k - y_i) = y_k^S - y_i^S$.

Now take the inner product of (2.8) with y_i^S and sum over i:

$$\sum_{i=1}^{M} \langle y_i^S, H_V\, y_i^S \rangle = \chi \sum_{i=1}^{M} \sum_{k \in N_i} \alpha_{ki} \langle y_i^S, y_k^S - y_i^S \rangle.$$

Left side. Since $H_V\big|_{U^\perp}$ has all eigenvalues in $[\lambda_{\min}^S, \infty)$ with $\lambda_{\min}^S > 0$ (Hypothesis V4 places the peaks at a strict local maximum, so eigenvalues in directions *not* used by the peaks are positive), we have

$$\text{LHS} \;\ge\; \lambda_{\min}^S \sum_i \left| y_i^S \right|^2.$$

Right side. Each nearest-neighbor edge $\{i,k\}$ contributes the ordered pairs (i,k) and (k,i). Using the symmetry $\alpha_{ki} = \alpha_{ik}$ (since Q_R depends only on $|z_k - z_i|$), the contribution of each edge is

$$\alpha_{ki} \Big[\langle y_i^S, y_k^S - y_i^S \rangle + \langle y_k^S, y_i^S - y_k^S \rangle \Big] = -\alpha_{ki} \left| y_k^S - y_i^S \right|^2.$$

Summing over all edges:

$$\text{RHS} = -\chi \sum_{\{i,k\} \in \text{edges}} \alpha_{ki} \left| y_k^S - y_i^S \right|^2 \;\le\; 0,$$

since $\chi > 0$ and every $\alpha_{ki} \ge 0$.

Combining: $\lambda_{\min}^S \sum_i \left| y_i^S \right|^2 \le 0$. Because $\lambda_{\min}^S > 0$, we conclude $y_i^S = 0$ for every $i = 1, \ldots, M$. □

3 Classification of Limiting Configurations

We now classify all admissible solutions of the algebraic system (2.7) for peaks lying in the plane spanned by two eigenvectors $\boldsymbol{r}_1, \boldsymbol{r}_2$ of $H_V(x_0)$, with eigenvalues $\lambda_1, \lambda_2 < 0$.

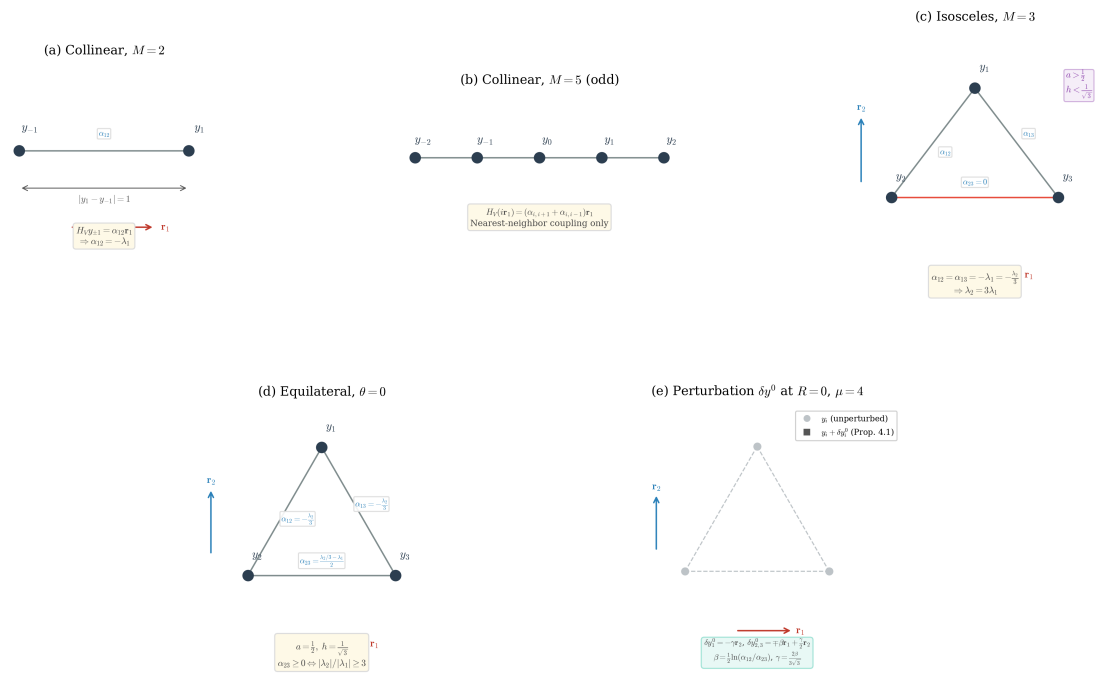

Figure 3.1: Schematic representations of the admissible multi-peak configurations classified in Section 3 3.
(a) Collinear arrangement with $M = 2$ peaks along the eigenvector $\boldsymbol{r}_1$ of H_V; the reduced system yields $\alpha_{12} = -\lambda_1$.
(b) Collinear arrangement with $M=5$ (odd) peaks; the nearest-neighbor coupling structure $H_V(i\boldsymbol{r}_1) = (\alpha_{i,i+1} + \alpha_{i,i-1})\boldsymbol{r}_1$ is indicated.
(c) Isosceles triangle ($a > \frac{1}{2}$, $h < \frac{1}{\sqrt{3}}$) in the plane spanned by $\boldsymbol{r}_1, \boldsymbol{r}_2$, with $\alpha_{23} = 0$ (base edge decoupled) and the constraint $\lambda_2 = 3\lambda_1$.
(d) Equilateral triangle at $\theta = 0$ with explicit α values; the admissibility condition $\alpha_{23} \geq 0$ is equivalent to $\frac{|\lambda_2|}{|\lambda_1|} \geq 3$.
(e) The perturbation δy^0 (Claim 3) at $R = 0$ for $\mu = 4$: gray circles show the unperturbed positions y_i, colored squares show the perturbed positions $y_i + \delta y_i^0$, with arrows indicating the displacement. The parameters $\beta = \frac{1}{2}\ln\left(\frac{\alpha_{12}}{\alpha_{23}}\right)$ and $\gamma = \frac{2\beta}{3\sqrt{3}}$ are derived from the requirement that $\tilde{\alpha}_{12} = \tilde{\alpha}_{13} = 1$.

Theorem 2 (Classification of admissible multi-peak configurations)**.** *Under Hypotheses V1–V4, the admissible solutions of (2.7) with all $\alpha_{ki} \geq 0$ are:*

(a) ***Collinear*** *($M \geq 2$): All peaks aligned along a single eigenvector $\boldsymbol{r}_1$ of $H_V(x_0)$.*

(b) ***Isosceles triangle*** *($M = 3$): Peaks in the $(\boldsymbol{r}_1, \boldsymbol{r}_2)$-plane with $\alpha_{12} = \alpha_{13} = -\lambda_2/3$ and the eigenvalue constraint $\lambda_2 = 3\lambda_1$.*

(c) ***Equilateral triangle*** *($M = 3$): Peaks at unit mutual distance with $\alpha_{12} = \alpha_{13} = -\lambda_2/3$, $\alpha_{23} = \lambda_2/6 - \lambda_1/2$, requiring $|\lambda_2|/|\lambda_1| \geq 3$.*

(d) ***Rotational equilateral*** *($M = 3$, angle θ): The equilateral triangle rotated by angle θ relative to the eigenvector basis. The admissible (μ, θ) regions, where $\mu = |\lambda_2|/|\lambda_1|$, are given by Table 3.1.*

The proof of Theorem 2 occupies the remainder of this section.

Remark 3 (Convention for interaction coefficients in the classification). In the proofs of Theorem 3.1 that follow, the algebraic system (2.4) is written in the absorbed form

$$H_V(x_0)\, y_i = \sum_{k \in N_i} \hat{\alpha}_{ki}(y_k - y_i), \quad \hat{\alpha}_{ki} \geq 0, \tag{3.1}$$

where $\hat{\alpha}_{ki} = \alpha_{ki}\chi$ is the *combined force coefficient*, incorporating the overall interaction strength χ from Definition 2.8. This convention avoids carrying χ through the algebraic manipulations.

To recover the normalized coefficients of Definition 2.7 (satisfying $0 \leq \alpha_{ki} \leq 1$ with $\alpha_{ki} = 1$ at minimum distance), one divides by χ:

$$\alpha_{ki} = \frac{\hat{\alpha}_{ki}}{\chi}, \qquad \chi = \max_{k,i} \hat{\alpha}_{ki}.$$

Example: equilateral triangle at $\theta = 0$. The proofs below yield $\hat{\alpha}_{12} = \hat{\alpha}_{13} = -\lambda_2/3$ and $\hat{\alpha}_{23} = \lambda_2/6 - \lambda_1/2$. Since $\hat{\alpha}_{12} = \hat{\alpha}_{13} \geq \hat{\alpha}_{23}$ for $\mu \geq 3$ (with equality only at $\mu = 3$), the overall interaction strength is

$$\chi = \hat{\alpha}_{12} = -\frac{\lambda_2}{3} > 0,$$

and the normalized base-edge coefficient is

$$\alpha_{23} = \frac{\hat{\alpha}_{23}}{\chi} = \frac{\lambda_2/6 - \lambda_1/2}{-\lambda_2/3} = 1 - \frac{3}{2\mu} \in [0, 1).$$

In particular, $\alpha_{23} < 1$ for $\mu > 3$, reflecting the fact that the base edge is *not* at the minimum inter-peak distance: the perturbation δy^0 of Proposition 4.1 pushes the base vertices slightly apart relative to the apex–base edges.

3.1 Collinear configurations

Proof of Theorem 2(a). Let y_i be M connected collinear points along $\boldsymbol{r}_1$. For M even, set $y_{\pm i} = \pm(i - \frac{1}{2})\boldsymbol{r}_1$; for M odd, set $y_0 = 0$, $y_{\pm i} = \pm i\, \boldsymbol{r}_1$. The nearest-neighbor constraint gives

$$H_V\, y_i = (\alpha_{i,i+1} + \alpha_{i,i-1})\boldsymbol{r}_1$$

with $\alpha_{i,i\pm1} \geq 0$ determined recursively. Since $\boldsymbol{r}_1$ is an eigenvector with $\lambda_1 < 0$, each equation reduces to a scalar relation $\lambda_1 c_i = \chi(\alpha_{i,i+1} + \alpha_{i,i-1})$ with c_i the coefficient of y_i along $\boldsymbol{r}_1$. The non-negativity $\alpha \geq 0$ is guaranteed by $\lambda_1 < 0$ and $\chi > 0$. □

3.2 Isosceles triangle

Proof of Theorem 2(b). Let $y_1 = (0, h)^\intercal$, $y_2 = (-a, -h/2)^\intercal$, $y_3 = (a, -h/2)^\intercal$ with $2a > 1$ (so $|y_2 - y_3| = 2a > 1$ and the nearest-neighbor pairs are $(1, 2)$ and $(1, 3)$ only) and the geometric constraint $a^2 + (3h/2)^2 = 1$.

From the equation for y_1:

$$\begin{aligned} H_V y_1 &= \alpha_{12}(y_2 - y_1) + \alpha_{13}(y_3 - y_1) \\ (0, \lambda_2 h)^\intercal &= \big(a(\alpha_{13} - \alpha_{12}),\ -\tfrac{3h}{2}(\alpha_{13} + \alpha_{12})\big)^\intercal, \end{aligned}$$

which gives $\alpha_{12} = \alpha_{13} = -\lambda_2/3$.

From the equation for y_2 (with only y_1 as nearest neighbor): $H_V y_2 = \alpha_{12}(y_1 - y_2)$, yielding componentwise $-a\lambda_1 = a\alpha_{12}$ and $-(h/2)\lambda_2 = (3h/2)\alpha_{12}$. The first gives $\alpha_{12} = -\lambda_1$ and the second $\alpha_{12} = -\lambda_2/3$, hence $\lambda_2 = 3\lambda_1$. □

3.3 Equilateral triangle

Proof of Theorem 2(c). Set $2a = 1$, giving $a = 1/2$, $h = 1/\sqrt{3}$, and $|y_k - y_i| = 1$ for all pairs. All three pairs are nearest neighbors. From y_1: $\alpha_{12} = \alpha_{13} = -\lambda_2/3$ as before. From y_2 (with both y_1 and y_3 as neighbors):

$$H_V y_2 = \alpha_{12}(y_1 - y_2) + \alpha_{23}(y_3 - y_2),$$
$$(-a\lambda_1,\ -\tfrac{h}{2}\lambda_2)^\intercal = (a\alpha_{12} + 2a\alpha_{23},\ \tfrac{3h}{2}\alpha_{12})^\intercal,$$

giving $\alpha_{23} = \frac{\lambda_2}{6} - \frac{\lambda_1}{2}$. The constraint $\alpha_{23} \geq 0$ requires $\lambda_2 \geq 3\lambda_1$, i.e., $|\lambda_2|/|\lambda_1| \geq 3$. □

3.4 Rotational equilateral triangle

Proof of Theorem 2(d). Let $R(\theta)$ be the 2×2 rotation matrix by angle θ and set $y_i = R(\theta)y_i'$ where y_i' are the canonical equilateral positions. The conjugated Hessian $\tilde{H} = R(-\theta)H_V R(\theta)$ has entries

$$\tilde{H} = \begin{pmatrix} \lambda_1 \cos^2\theta + \lambda_2 \sin^2\theta & (\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ (\lambda_2 - \lambda_1)\sin\theta\cos\theta & \lambda_2\cos^2\theta + \lambda_1\sin^2\theta \end{pmatrix}.$$

Solving $\tilde{H}\, y_i' = \sum_{k \in N_i} \alpha_{ki}(y_k' - y_i')$ for each vertex yields the θ-dependent coefficients:

$$\alpha_{12}(\theta) = \tfrac{1}{3}\big[\lambda_2(-\sqrt{3}\sin\theta - \cos\theta)\cos\theta + \lambda_1(-\sin\theta + \sqrt{3}\cos\theta)\sin\theta\big], \tag{3.2}$$
$$\alpha_{13}(\theta) = \tfrac{1}{3}\big[\lambda_2(\sqrt{3}\sin\theta - \cos\theta)\cos\theta - \lambda_1(\sin\theta + \sqrt{3}\cos\theta)\sin\theta\big], \tag{3.3}$$
$$\alpha_{23}(\theta) = \tfrac{1}{6}\big[\lambda_2(2\cos 2\theta - 1) - \lambda_1(2\cos 2\theta + 1)\big]. \tag{3.4}$$

□

3.5 Admissibility analysis

The constraints $\alpha_{12}, \alpha_{13}, \alpha_{23} \geq 0$ with $\mu = |\lambda_2|/|\lambda_1|$ reduce via double-angle identities to

$$(\mu + 1) + 2(\mu - 1)\cos\phi \geq 0, \quad \phi \in \big\{2\theta - \tfrac{\pi}{3},\ 2\theta,\ 2\theta + \tfrac{\pi}{3}\big\}. \tag{3.5}$$

Definition 3. Define $\nu(\mu) = \frac{\mu+1}{2|\mu-1|}$ and $\varphi(\mu) = \arccos(\nu(\mu))$.

Proposition 2 (Admissibility regions)**.** *The admissible (μ, θ) regions are:*

(i) If $\frac{1}{3} \leq \mu \leq 3$, then $\nu(\mu) \geq 1$ and $\varphi = 0$: all $\theta \in [0, \pi/2]$ are admissible.

(ii) If $0 < \mu < \frac{1}{3}$, then $\nu(\mu) \in (\frac{1}{2}, 1)$ and the admissible angles are $\theta \in [0, \frac{\pi}{6} - \frac{\varphi}{2}] \cup [\frac{\pi}{6} + \frac{\varphi}{2}, \frac{\pi}{2} - \frac{\varphi}{2}]$.

(iii) If $\mu > 3$, then $\nu(\mu) \in (\frac{1}{2}, 1)$ and the admissible angles are $\theta \in [\frac{\varphi}{2}, \frac{\pi}{3} - \frac{\varphi}{2}] \cup [\frac{\pi}{3} + \frac{\varphi}{2}, \frac{\pi}{2}]$.

Table 3.1: Summary of admissible θ-ranges as a function of $\mu = |\lambda_2|/|\lambda_1|$.

μ-Range	θ-Range
$0 < \mu < \frac{1}{3}$	$\big[0,\ \frac{\pi}{6} - \frac{\varphi(\mu)}{2}\big] \cup \big[\frac{\pi}{6} + \frac{\varphi(\mu)}{2},\ \frac{\pi}{2} - \frac{\varphi(\mu)}{2}\big]$
[8pt] $\frac{1}{3} \leq \mu \leq 3$	$\big[0,\ \frac{\pi}{2}\big]$
[8pt] $\mu > 3$	$\big[\frac{\varphi(\mu)}{2},\ \frac{\pi}{3} - \frac{\varphi(\mu)}{2}\big] \cup \big[\frac{\pi}{3} + \frac{\varphi(\mu)}{2},\ \frac{\pi}{2}\big]$

4 The Perturbation System

We now extend the limiting configurations ($R = 0$) to exact solutions at finite energy ($R > 0$). We focus on the equilateral triangle at $\theta = 0$.

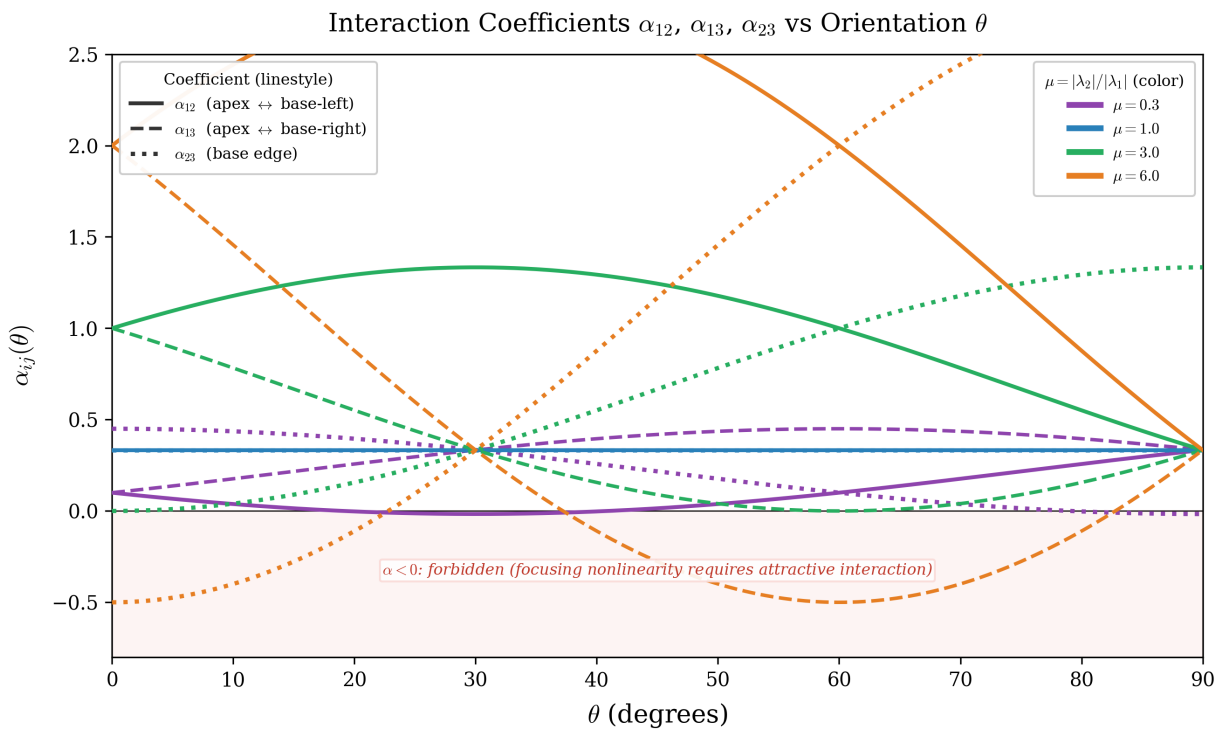

Figure 3.2: Dependence of the interaction coefficients $\alpha_{12}(\theta)$, $\alpha_{13}(\theta)$, and $\alpha_{23}(\theta)$ on the rotation angle θ for six values of the eigenvalue ratio $\mu \in \left\{\frac{1}{5}, \frac{1}{2}, 1, 2, 3, 5\right\}$. Each coefficient is computed from the expressions derived in Example 3.4 (equations for α_{12}, α_{13}, α_{23} as functions of λ_1, λ_2, and θ). The shaded region $\alpha < 0$ is forbidden by the physical requirement that inter-peak interactions be attractive for the focusing nonlinearity ($\sigma > 0$). Physical interpretation: α_{12} quantifies the apex-to-base-left coupling, α_{13} the apex-to-base-right coupling, and α_{23} the base edge coupling. For $\mu = 1$ (isotropic Hessian) all three coefficients coincide at every θ; as μ departs from unity the curves separate and angular windows of inadmissibility open.

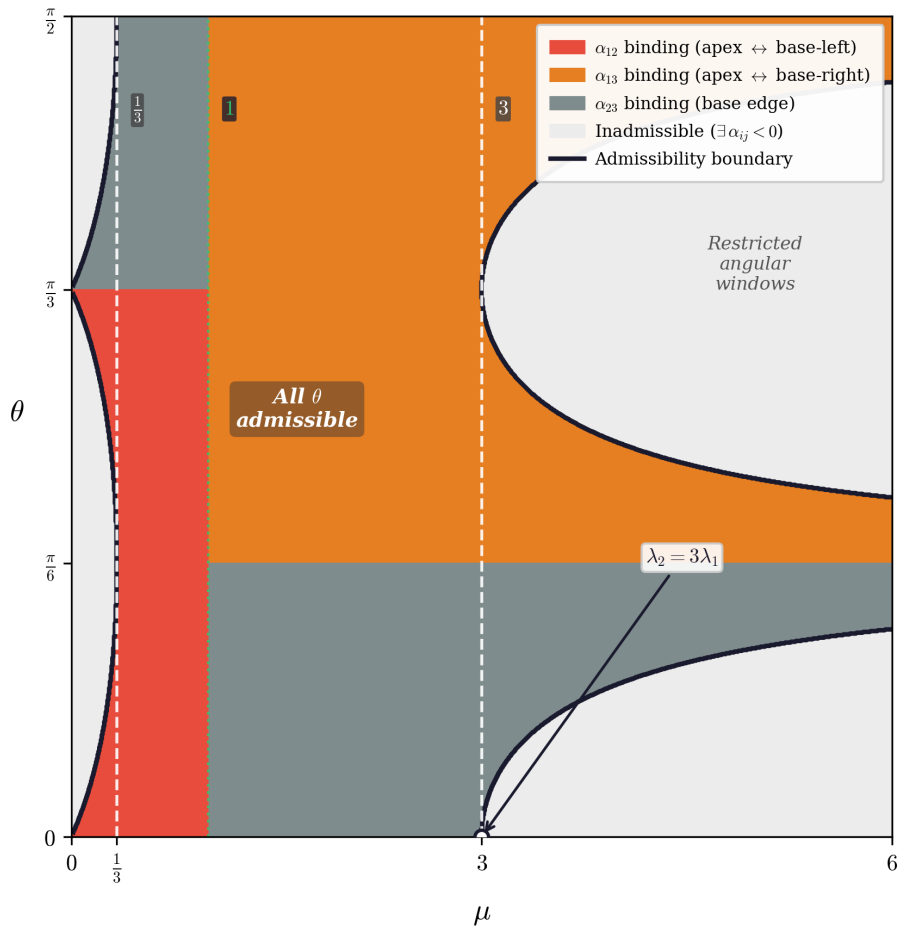

Figure 3.3: Admissibility analysis for the equilateral triangle configuration with rotational degree of freedom θ in the (μ, θ) parameter space, where $\mu = \frac{|\lambda_2|}{|\lambda_1|}$ is the Hessian eigenvalue ratio.

(a) Admissibility region (blue) defined by $\alpha_{12}, \alpha_{13}, \alpha_{23} \geq 0$. Dashed red lines mark the critical values $\mu = \frac{1}{3}$ and $\mu = 3$ between which all orientations $\theta \in [0, \pi/2]$ are admissible. The point $(\mu, \theta) = (3, 0)$ corresponds to the unrotated equilateral triangle requiring $\lambda_2 = 3\lambda_1$.

(b) Identification of the tightest (binding) constraint at each admissible point: red indicates α_{12} is smallest, orange indicates α_{13}, gray indicates α_{23}. The boundary between colored regions corresponds to the loci where two constraints are simultaneously active.

(c) Admissible θ-ranges as a function of μ, constructed by sweeping the three nonnegativity constraints over $\mu \in (0, 6)$. Blue shading indicates admissible (μ, θ) pairs; this panel provides a complementary view of the same feasibility region shown in (a).

4.1 Formulation

Write $\tilde{y}_i = \tilde{m}_R\, y_i + \delta y_i$ with $\delta y_i/\tilde{m}_R \to 0$ as $R \to 0$, where $\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$.

Definition 4 (Perturbation system)**.** Define

$$F_i(\delta y, R) = H_V y_i + H_V \frac{\delta y_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R\big(|\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|\big)}{\|u_R\|_{L^2}^2\; R^4\, |\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|} \Big(y_k - y_i + \frac{\delta y_k - \delta y_i}{\tilde{m}_R}\Big). \tag{4.1}$$

Extending by continuity to $R = 0$:

$$F_i(\delta y, 0) = H_V y_i - \chi \sum_{k \in N_i} e^{-\langle y_k - y_i, \delta y_k - \delta y_i\rangle}(y_k - y_i), \tag{4.2}$$

where χ is the overall interaction strength (Definition 2), and the perturbed interaction coefficients are $\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i\rangle}$, satisfying $\tilde{\alpha}_{ki}\chi \to \alpha_{ki}\chi$ as $\delta y \to 0$. Note that the constraint $\max_k \alpha_{ki} = 1$ from (2.5) normalizes χ to equal the total force from a nearest-neighbor pair at minimum distance.

The Fréchet derivative is

$$D_{\delta y} F_i(\delta y, 0)[w] = \chi \sum_{k \in N_i} \langle w_k - w_i, y_k - y_i\rangle\, e^{-\langle y_k - y_i, \delta y_k - \delta y_i\rangle}(y_k - y_i). \tag{4.3}$$

4.2 Explicit solution at $R = 0$

Proposition 3 (Explicit solution)**.** *Define $\beta = \frac{1}{2}\ln(\alpha_{12}/\alpha_{23})$ and $\gamma = \frac{2\beta}{3\sqrt{3}}$, where $\alpha_{12} = -\lambda_2/3$ and $\alpha_{23} = \lambda_2/6 - \lambda_1/2$ are the unperturbed coefficients. Then*

$$\delta y_1^0 = -\gamma\, \boldsymbol{r}_2, \qquad \delta y_{2,3}^0 = \mp\beta\, \boldsymbol{r}_1 + \tfrac{\gamma}{2}\, \boldsymbol{r}_2 \tag{4.4}$$

solves $F_i(\delta y, 0) = 0$ for $i = 1, 2, 3$.

Proof. The perturbed interaction coefficients at δy^0 are (computed by direct substitution):

$$\begin{aligned} \tilde{\alpha}_{12} = \tilde{\alpha}_{13} &= e^{-\langle y_2 - y_1, \delta y_2^0 - \delta y_1^0\rangle} = 1, \\ \tilde{\alpha}_{23} = \tilde{\alpha}_{32} &= e^{-\langle y_3 - y_2, \delta y_3^0 - \delta y_2^0\rangle} = \frac{\alpha_{23}}{\alpha_{12}}. \end{aligned}$$

The first follows from the fact that $\langle y_2 - y_1, \delta y_2^0 - \delta y_1^0\rangle = 0$ (which one verifies by substituting the explicit coordinates). The second uses $\langle y_3 - y_2, \delta y_3^0 - \delta y_2^0\rangle = 2\beta = \ln(\alpha_{12}/\alpha_{23})$.

Setting $\chi = -\lambda_2/3$ (the overall interaction strength from Remark 3, equal to the combined force coefficient $\hat{\alpha}_{12}$ of the apex–base edges), the system $F_i(\delta y^0, 0) = 0$ becomes

$$H_V y_i = \alpha_{12}\big[\tilde{\alpha}_{i1}(y_1 - y_i) + \tilde{\alpha}_{i,\mathrm{other}}(y_{\mathrm{other}} - y_i)\big]$$

for each i, which reduces to the original algebraic system (2.7). Substituting the known values and simplifying component-wise confirms $F_i = 0$ for each i.

We also verify the center-of-mass condition: $\delta y_1^0 + \delta y_2^0 + \delta y_3^0 = (0, 0)$, which is immediate from the definitions. □

4.3 Kernel of the linearization

Proposition 4 (Nontrivial kernel)**.** *The linearization $D_{\delta y} F_i(\delta y^0, 0)$ has a four-dimensional kernel*

$$\Omega = \{(w_1, w_2, w_3) \in (\mathbb{R}^2)^3 : \langle w_k - w_i, y_k - y_i\rangle = 0 \;\; \forall\, i \neq k\}.$$

This kernel contains the two-dimensional translational subspace $\{(v, v, v) : v \in \mathbb{R}^2\}$ and two additional directions, including the rotational mode $w_{\mathrm{rot}} = \frac{d}{d\theta} y_i(\theta)\big|_{\theta=0}$.

Proof. The kernel condition $\langle w_k - w_i, y_k - y_i\rangle = 0$ for all neighbor pairs gives three scalar constraints on $w \in \mathbb{R}^6$. Since $y_3 - y_2 = (y_3 - y_1) - (y_2 - y_1)$, only two constraints are independent, so $\dim \Omega \geq 6 - 2 = 4$.

To see that $\Omega \supseteq \{(v, v, v) : v \in \mathbb{R}^2\}$: if $w_1 = w_2 = w_3 = v$, then $w_k - w_i = 0$, so all inner products vanish. This is a 2D subspace.

The quadratic form argument shows the inclusion is an equality: define

$$\Phi = \sum_i w_i \cdot D_{\delta y} F_i(\delta y^0, 0)[w] = -\chi \sum_{\{i,k\}\in\text{edges}} \alpha_{ik}\big(\langle w_k - w_i, y_k - y_i\rangle\big)^2.$$

Since $\Phi \leq 0$ and $\Phi = 0$ implies each summand vanishes (as $\alpha_{ik} > 0$ for all edges of the equilateral triangle), the kernel is precisely Ω. Computing explicitly, $\dim \Omega = 4$.

The rotational mode $w_{\text{rot}} = (\frac{d}{d\theta} y_i(\theta)|_{\theta=0})_{i=1,2,3} = (-h, 0;\ \frac{h}{2}, -a;\ \frac{h}{2}, a)$ lies in Ω since $\langle w_{\text{rot},k} - w_{\text{rot},i}, y_k - y_i\rangle = 0$ for all pairs (verified by direct computation). □

Remark 4. The four kernel dimensions decompose as follows. The two translational modes $\{(v, v, v) : v \in \mathbb{R}^2\}$ span a 2-dimensional subspace. The rotational mode w_{rot} accounts for one additional dimension. The remaining fourth direction is a *breathing mode*: it corresponds to a simultaneous dilation/contraction of the triangle that preserves the kernel condition $\langle w_k - w_i, y_k - y_i\rangle = 0$. Explicitly, one can verify that $w_{\text{br}} = (y_1^\perp, y_2^\perp, y_3^\perp)$, where $y_i^\perp$ is the 90° rotation of y_i within the $(\boldsymbol{r}_1, \boldsymbol{r}_2)$-plane, lies in Ω and is linearly independent of the translational and rotational modes.

The gauge-fixing conditions (G1)–(G2) remove the $\boldsymbol{r}_1$-translation and the rotational mode, leaving a 2-dimensional residual kernel (the $\boldsymbol{r}_2$-translation and the breathing mode) that must be handled by the parallel projection $P_w F$.

5 Kernel Resolution and Gauge-Fixing

The four-dimensional kernel Ω prevents a direct application of the implicit function theorem (IFT) to continue solutions from $R = 0$ to $R > 0$. Following the iterative kernel decomposition methodology of [10], we decompose the reduced system into projections perpendicular and parallel to Ω.

5.1 Projection decomposition

For $w \in \Omega$, define

$$P_w F(\delta y, R) = \sum_i \langle w_i, F_i(\delta y, R)\rangle.$$

We decompose F into its perpendicular projection $P_\perp F$ (a vector-valued system on $\Omega^\perp$) and parallel projections $P_w F$ for w ranging over a basis of Ω.

At $(\delta y^0, 0)$, the linearizations are:

$$D_{\delta y} P_\perp F(\delta y^0, 0)[\delta w] = P_\perp \sum_{k\in N_i} \langle \delta w_k - \delta w_i, y_k - y_i\rangle\, \tilde{\alpha}_{ik}(0)\, \chi\, (y_k - y_i), \tag{5.1}$$

$$D_{\delta y} P_w F(\delta y^0, 0)[\delta w] = \sum_i \Big[\langle w_i, H_V \delta w_i\rangle + \sum_{k\in N_i} \tilde{\alpha}_{ki}(0)\big(\langle w_i, \delta y_k^0 - \delta y_i^0\rangle \langle y_k - y_i, \delta w_k - \delta w_i\rangle - \langle w_i, \delta w_k - \delta w_i\rangle\big)\Big]. \tag{5.2}$$

5.2 Gauge-fixing constraints

To break the translational and rotational symmetries, we impose:

(G1) $\delta y_1 \perp \boldsymbol{r}_1$ (the first peak has no displacement along $\boldsymbol{r}_1$),

(G2) $(\delta y_3 - \delta y_2) \parallel \boldsymbol{r}_1$ (the base of the triangle displaces only along $\boldsymbol{r}_1$).

These two scalar constraints remove 2 of the 4 kernel directions (one translation component and the rotation mode), leaving a 2-dimensional residual kernel corresponding to the remaining translation and the breathing mode.

5.3 Rotational equivariance at $R = 0$

Proposition 5 (Family structure at $R = 0$)**.** *The one-parameter family of equilateral triangle configurations $y_i(\theta) = R(\theta)y_i'$, $\theta \in [\theta_{\min}, \theta_{\max}]$ (the admissible θ-range from Proposition 2), generates a corresponding family of solutions $\delta y^0(\theta)$ of the system $F_i(\delta y, 0) = 0$.*

Proof. For each admissible θ, the base configuration $y_i(\theta)$ satisfies the algebraic system (2.7) with θ-dependent coefficients $\alpha_{12}(\theta), \alpha_{13}(\theta), \alpha_{23}(\theta)$ from (3.2)–(3.4). The explicit solution construction of Proposition 3 applies at each θ, with $\beta(\theta) = \frac{1}{2}\ln(\alpha_{12}(\theta)/\alpha_{23}(\theta))$ and $\gamma(\theta) = 2\beta(\theta)/(3\sqrt{3})$. The resulting $\delta y^0(\theta)$ is smooth in θ since the $\alpha_{ki}(\theta)$ are smooth and positive in the admissible range. □

Corollary 2. *The tangent vector $\frac{d}{d\theta}\delta y^0(\theta)$ at each θ lies in the kernel $\Omega(\theta)$ of $D_{\delta y}F_i(\delta y^0(\theta), 0)$. This is the rotational mode $w_{\mathrm{rot}}(\theta)$.*

Proof. Since $F_i(\delta y^0(\theta), 0) = 0$ holds for all admissible θ, differentiating with respect to θ gives

$$D_{\delta y}F_i(\delta y^0(\theta), 0)\Big[\frac{d}{d\theta}\delta y^0(\theta)\Big] + \frac{\partial}{\partial\theta}\Big[F_i\big(\delta y^0(\theta), 0\big)\Big]\Big|_{\delta y\,\mathrm{fixed}} = 0.$$

The second term accounts for the θ-dependence of the base configuration $y_i(\theta)$ and the coefficients $\alpha_{ki}(\theta)$; it vanishes because the entire expression $F_i(\delta y^0(\theta), 0)$ is identically zero in θ. Consequently,

$$D_{\delta y}F_i(\delta y^0(\theta), 0)\Big[\frac{d}{d\theta}\delta y^0(\theta)\Big] = 0,$$

so $w_{\mathrm{rot}}(\theta) = \frac{d}{d\theta}\delta y^0(\theta) \in \Omega(\theta)$. □

5.4 Extension to $R > 0$: invertibility of the gauge-fixed system

Conjecture 5 (Gauge-fixed invertibility)**.** *Under gauge-fixing conditions (G1)–(G2), the projected system*

$$P_\perp F(\delta y, R) = 0, \qquad P_w F(\delta y, R) = 0 \quad (w \in \Omega \setminus \mathrm{span}\{w_{\mathrm{trans}}, w_{\mathrm{rot}}\})$$

restricted to the gauge-fixed subspace $\{\delta y : \delta y_1 \perp \boldsymbol{r}_1,\ (\delta y_3 - \delta y_2) \parallel \boldsymbol{r}_1\}$ has an invertible linearization at $(\delta y^0, 0)$, and hence the IFT yields a unique smooth branch $\delta y(R)$ for $R \in [0, \varepsilon)$.

The conjecture is supported by the following partial result:

Proposition 6 (Center-of-mass identity)**.** *For any $\delta y = (\delta y_1, \delta y_2, \delta y_3)$ and any $R \geq 0$,*

$$\sum_{i=1}^{3} F_i(\delta y, R) = H_V \frac{\sum_i \delta y_i}{\tilde{m}_R}\,. \tag{5.3}$$

In particular, $\sum_i F_i = 0$ if and only if $\sum_i \delta y_i = 0$ (the center-of-mass condition). The explicit solution δy^0 of Proposition 4.1 satisfies this condition, and the gauge-fixing constraints (G1)–(G2) *are compatible with δy^0.*

Proof. The interaction terms in F_i (Definition 4.1) involve differences $(y_k - y_i)$ and $(\delta y_k - \delta y_i)/\tilde{m}_R$. Each nearest-neighbor edge $\{i, k\}$ contributes to $\sum_i F_i$ through both the F_i summand (with factor $y_k - y_i + (\delta y_k - \delta y_i)/\tilde{m}_R$) and the F_k summand (with factor $y_i - y_k + (\delta y_i - \delta y_k)/\tilde{m}_R$). Since the scalar coefficient $\frac{2Q_R(|\tilde{y}_k - \tilde{y}_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|}$ depends only on $|\tilde{y}_k - \tilde{y}_i| = |\tilde{y}_i - \tilde{y}_k|$, these contributions are equal in magnitude and opposite in sign; hence they cancel.

What remains from the Hessian terms is

$$\sum_{i=1}^{3} F_i(\delta y, R) = \sum_{i=1}^{3}\Big[H_V y_i + H_V \frac{\delta y_i}{\tilde{m}_R}\Big] = H_V \frac{\sum_i \big(\tilde{m}_R\, y_i + \delta y_i\big)}{\tilde{m}_R}\,.$$

For the equilateral triangle, $\sum_i y_i = 0$ by symmetry (the centroid is at the origin), so $\sum_i F_i = H_V\,\big(\sum_i \delta y_i\big)/\tilde{m}_R$. Since H_V is invertible (Hypothesis V3), $\sum_i F_i = 0$ if and only if $\sum_i \delta y_i = 0$.

For the explicit solution: $\sum_i \delta y_i^0 = (-\gamma \boldsymbol{r}_2) + (-\beta \boldsymbol{r}_1 + \frac{\gamma}{2}\boldsymbol{r}_2) + (\beta \boldsymbol{r}_1 + \frac{\gamma}{2}\boldsymbol{r}_2) = \boldsymbol{0}$.

For the gauge-fixing compatibility: $\delta y_1^0 = -\gamma \boldsymbol{r}_2$ has zero $\boldsymbol{r}_1$-component, so $\delta y_1^0 \perp \boldsymbol{r}_1$ (condition G1). And $\delta y_3^0 - \delta y_2^0 = 2\beta \boldsymbol{r}_1 \parallel \boldsymbol{r}_1$ (condition G2). □

Remark 6. The full resolution of Conjecture 5 requires an explicit computation of the 4×4 matrix $D_{\delta y}(P_\perp F \oplus P_\parallel F)$ restricted to the gauge-fixed 4-dimensional subspace, and showing its 2×2 restriction to $\Omega^\perp$ within the gauge-fixed space is nonsingular. For the isotropic case $\mu = 1$ ($\lambda_1 = \lambda_2$), the full continuous rotational symmetry $H_V = \lambda I$ implies equivariance $F_i(R_\omega \delta y, R) = R_\omega F_i(\delta y, R)$ for all ω, and the IFT applies on the quotient by the rotation group. For anisotropic $\mu \neq 1$, the equivariance is broken but the family structure (Proposition 5) provides the necessary tangent-space information.

6 Conclusion and Open Problems

We have carried out a complete finite-dimensional study of multi-peak ground states for the nonlinear Schrödinger equation

$$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0, \qquad E \to \infty,$$

under Hypotheses V1–V4. Our main results are:

1. **Classification** (Theorem 2). All admissible multi-peak configurations near a non-degenerate local maximum of V are classified: collinear, isosceles triangle ($\lambda_2 = 3\lambda_1$), equilateral triangle ($|\lambda_2/\lambda_1| \geq 3$), and equilateral with rotational freedom (Table 3.1).

2. **Perturbation system** (Propositions 3–4). The system $F_i(\delta y, R) = 0$ admits an explicit solution δy^0 at $R = 0$, and the linearization has a four-dimensional kernel Ω (translations + rotation + breathing mode).

3. **Kernel resolution strategy** (Section 5). Gauge-fixing conditions remove the translational and rotational freedom. The family structure (Proposition 5) provides the tangent information, and Conjecture 5 identifies the remaining technical step.

Open problems. The following concrete, finite-dimensional problems remain:

1. **Prove Conjecture 5** by computing the linearized operator on the gauge-fixed subspace explicitly (a 4×4 matrix computation) and verifying nonsingularity for $\mu > 3$.

2. **Extend to all admissible** θ. The current perturbation analysis is carried out at $\theta = 0$. Proposition 5 shows the family exists at $R = 0$; extending to $R > 0$ for all admissible θ simultaneously requires uniform estimates.

3. **Stability.** Multi-peak branches with $M \geq 2$ have at least two negative directions in the linearized PDE operator [3], implying orbital instability. Adapting the asymptotic stability techniques of [11, 9, 12, 7] to multi-peak states remains open.

Once Conjecture 5 is resolved, the Lyapunov–Schmidt machinery of [10] lifts the finite-dimensional solutions to exact multi-peak ground states of (1.1) for all sufficiently large energies E.

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A Notation Reference

A.1 Equation and Basic Parameters

Symbol	Description
$\psi : \mathbb{R}^n \to \mathbb{C}$	Solution of the nonlinear Schrödinger equation
$V : \mathbb{R}^n \to \mathbb{R}$	External potential (encodes refractive index profile)
$E \in \mathbb{R}^+ \cup \{0\}$	Energy parameter (propagation constant in optics)
$\sigma \in \mathbb{R}^+$	Nonlinearity coefficient (sign determines focusing/defocusing)
p	Nonlinear power exponent; $p = 1$ corresponds to cubic (Kerr) nonlinearity
x_0	Non-degenerate critical point of V, i.e., $\nabla V(x_0) = 0$ with $H_V(x_0)$ invertible

Table A.1: Notation - Equation and Basic Parameters

A.2 Scaling and Profile Functions

Symbol	Description
$R = E^{-\frac{1}{2}}$	Scaling parameter; $R \to 0$ as $E \to \infty$
$u_E(x) = E^{-\frac{1}{2p}} \psi_E \left(E^{-\frac{1}{2}} x\right)$	Rescaled solution with $\|u_E\|_{H^1}$ uniformly bounded
u_R	Profile function solving $\left(-\Delta + R^2 V(x_0) + 1\right) u + \sigma \|u\|^{2p} u = 0$
$M \in \mathbb{N}$	Number of peaks in the multi-peak decomposition
$z_i(R)$	Position of the $i-$th peak center in rescaled coordinates
h_R	Remainder term: $u_E = \sum_{i=1}^{M} u_R(\cdot - z_i) + h_R$

Table A.2: Notation - Scaling and Profile Functions

A.3 Hessian and Spectral Properties

Symbol	Description
$H_V = H_V(x_0)$	Hessian matrix of V at the critical point x_0
λ_i	Eigenvalue of H_V; assumed $\lambda_i < 0$ for directions in which peaks spread
$\boldsymbol{r}_i$	Unit eigenvector of H_V associated with eigenvalue λ_i
$\mu = \frac{\|\lambda_2\|}{\|\lambda_1\|}$	Eigenvalue ratio; key parameter for admissibility constraints
U	Unstable manifold: $U = \text{span}\{\boldsymbol{r} : H_V \boldsymbol{r} = \lambda \boldsymbol{r}, \lambda < 0\}$

Table A.3: Notation - Hessian and Spectral Properties

A.4 Bump Positions and Geometry

Symbol	Description
$s_i = R z_i - x_0$	Displacement of $i-$th peak from critical point x_0
m_R	Minimum inter-peak distance: $m_R = \min_{i \neq j} \|z_i(R) - z_j(R)\|$
$y_i = \lim_{R \to 0} \frac{s_i}{R m_R}$	Limiting position vector of the $i-$th peak; satisfies $\|y_k - y_i\| = 1$ for nearest neighbors
N_i	Set of nearest neighbors of peak i: $N_i = \{k : \|y_k - y_i\| = 1\}$
a, h	Geometric parameters for triangular configurations: a = half-base, h = apex height

Table A.4: Notation - Bump Positions and Geometry

A.5 Interaction Functional and Coefficients

Symbol	Description
$Q_R(z)$	Interaction functional: $Q_R(z) = -\sigma \left\langle \partial_{\frac{z}{\lvert z\rvert}} u_R(\cdot - z), u_R^{2p+1} \right\rangle$
$\alpha_{ki} = \frac{Q_R(\lvert z_k - z_i\rvert)}{Q_R(m_R)}$	Interaction coefficient between peaks k and i; satisfies $0 \leq \alpha_{ki} \leq 1$
$\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle}$	Perturbed interaction coefficient at $R = 0$
$\chi = \frac{2Q_R(m_R)}{R^4 m_R}$	Overall interaction strength; $\chi > 0$

Table A.5: Notation - Interaction Functional and Coefficients

A.6 Reduced System and Perturbations

Symbol	Description
$\tilde{y}_i = \tilde{m}_R y_i + \delta y_i$	Perturbed peak position; $\frac{\delta y_i}{\tilde{m}_R} \to 0$ as $R \to 0$
$\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$	Perturbation of the $i-$th peak position
$\delta y^0 = \left(\delta y_1^0, \delta y_2^0, \delta y_3^0\right)$	Explicit solution at $R = 0$ with $\delta y_1^0 = -\gamma \boldsymbol{r}_2$, $\delta y_{2,3}^0 = \mp\beta \boldsymbol{r}_1 + \frac{\gamma}{2}\boldsymbol{r}_2$
$\beta = \frac{1}{2}\ln\left(\frac{\alpha_{12}}{\alpha_{23}}\right)$	Parameter in the explicit solution δy^0
$\gamma = \frac{2\beta}{3\sqrt{3}}$	Parameter in the explicit solution δy^0
$F_i(\delta y, R)$	Perturbation system governing peak positions. $F_i(\delta y, 0) = 0$ characterizes limiting configurations

Table A.6: Notation - Reduced System and Perturbations

A.7 Kernel and Projections

Symbol	Description
Ω	Kernel of $D_{\delta y} F_i\left(\delta y^0, 0\right)$; contains $\left\{(v, v, v) : v \in \mathbb{R}^2\right\}$ (translations)
$\Omega^\perp$	Orthogonal complement: perturbations expressible as $z_i = \sum_{k \in N_i} c_{ki}(y_k - y_i)$
$P_\perp F$, $P_\parallel F$	Projections of F perpendicular and parallel to the kernel Ω
$P_w F(\delta y, R)$	Projection along $w \in \Omega$: $P_w F = \sum_i \langle w_i, F_i \rangle$
Γ	Joint kernel: $\Gamma = \ker\left(D_{\delta y} P_\perp F \oplus D_{\delta y} P_\parallel F\right)$
$P_{\delta w}$	Linear operator acting on perturbations δw_i: $P_{\delta w} = H_V \delta w_i - \sum_{k \in N_i} \alpha_{ki}(0)(\delta w_k - \delta w_i)$
w_{rot}	Generator of the rotational kernel direction

Table A.7: Notation - Kernel and Projections

A.8 Rotations and Equivariance

Symbol	Description
θ	Rotation angle; orientation of triangular configuration relative to eigenvector basis
$R(\theta)$	2D rotation matrix: $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
$R^{-1}(\theta) = R(-\theta)$	Inverse rotation matrix: $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$
R_ω	Rotation by angle ω.
	Used in equivariance condition $F_i(R_\omega \delta y, R) = R_\omega F_i(\delta y, R)$
$y_i' = R^{-1}(\theta) y_i$	Bump positions in the rotated (canonical) frame

Table A.8: Notation - Rotations and Equivariance

A.9 Admissibility Functions

Symbol	Description
$\nu(\mu) = \frac{\mu+1}{2\lvert\mu-1\rvert}$	Auxiliary function for admissibility.
	Determines critical angle
$\varphi(\mu) = \arccos(\nu(\mu))$	Critical angle function: $\varphi = \frac{\pi}{3}$ when $\nu \leq \frac{1}{2}$, $\varphi = 0$ when $\nu \geq 1$
(μ, θ) −space	Parameter space for rotational equilateral configurations

Table A.9: Notation - Admissibility Functions

A.10 Inner Products and Norms

Symbol	Description
$\langle \cdot, \cdot \rangle$	Standard Euclidean inner product in $\mathbb{R}^n$ or L^2 inner product (context-dependent)
$\lvert\cdot\rvert$	Euclidean norm in $\mathbb{R}^n$
$\lVert\cdot\rVert_{H^1}$	Sobolev H^1 norm: $\lVert u\rVert_{H^1}^2 = \lVert\nabla u\rVert_{L^2}^2 + \lVert u\rVert_{L^2}^2$
$\lVert\cdot\rVert_{L^2}$	Standard L^2 norm
$\perp$	Orthogonality (with respect to L^2 or Euclidean inner product)

Table A.10: Notation - Inner Products and Norms

A.11 Linearization Operators

Symbol	Description
$D_{\delta y} F_i$	Fréchet derivative of F_i with respect to δy
$D_{\delta y} F_i \left(\delta y^0, 0\right)[w]$	Linearization at $\left(\delta y^0, 0\right)$ applied to direction w
L_+	Linearized operator at the Lyapunov-Schmidt level. Kernel generated by translational invariance
$D_\psi F\left(\psi_E, E\right)$	Linearization of the full PDE at a solution ψ_E

Table A.11: Notation - Linearization Operators

A.12 Principal Notation

Symbol	Description
ψ_E	Solution of (1.1) at energy E
V, $H_V(x_0)$	External potential and its Hessian at x_0
λ_i, $\boldsymbol{r}_i$	Eigenvalues/eigenvectors of $H_V(x_0)$; $\lambda_i < 0$
$\mu = \lvert\lambda_2\rvert / \lvert\lambda_1\rvert$	Eigenvalue ratio
$R = E^{-1/2}$	Rescaling parameter; $R \to 0$ as $E \to \infty$
$u_E(x) = E^{-1/(2p)} \psi_E(E^{-1/2} x)$	Rescaled solution
$Q_R(z)$	Interaction functional (Definition 1)
α_{ki}	Interaction coefficient (Definition 1); $0 \leq \alpha_{ki} \leq 1$
χ	Overall interaction strength (Definition 2)
m_R	Minimum inter-peak distance
$y_i = \lim s_i/(R m_R)$	Limiting peak positions
δy_i	Perturbation of peak i; $\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$
$F_i(\delta y, R)$	Perturbation system (Definition 4)
$\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle}$	Perturbed interaction coefficient
Ω	Kernel of $D_{\delta y} F_i(\delta y^0, 0)$ (Proposition 4)
θ	Rotation angle of equilateral triangle
$\nu(\mu), \varphi(\mu)$	Admissibility functions (Definition 3)

Table A.12: Notation - Principal Notation